

Mateusz Markowski

AI Engineer · agentic systems, MCP & LLM orchestration · Fullstack (Kotlin / React)

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PROFILE

AI Engineer with production GenAI deployments in enterprise (insurance sector): delivered Assistance AI — a GenAI platform for 1,000+ users — in 11 days, an application recognized in the Rzeczpospolita Cyfrowa 2024 awards; co-building an MCP (Model Context Protocol) server compliant with the company security policy, plus agent evaluation and verification processes. Fullstack (Kotlin/Spring, React/TypeScript) is my toolset — from multi-agent systems and agentic workflows, through LLM orchestration, to RAG, embeddings and vector databases.

SKILLS

Agentic systems & MCP	Multi-agent systems — orchestrator, specialized subagents, ReAct pattern · MCP (Model Context Protocol) — designing servers compliant with security policies · Agentic workflows — task decomposition, routing, result aggregation
Production LLM integration	Claude (Claude Code), GPT / OpenAI API — LLM integrations in enterprise apps · LLM orchestration — combining models, tools and data into coherent flows · AI-SDLC: GitHub Copilot, Copilot Extensions — rollouts and workshops
Agent evaluation & testing	Agent evaluation, automated PR code review, response-quality assessment
RAG & data	RAG, embeddings, vector databases
Backend	Kotlin, Spring Boot / WebFlux, REST API, SQL, RabbitMQ, microservice architecture
Frontend	React.js, TypeScript, JavaScript
DevOps & QA	GitHub / Bitbucket, CI/CD, Selenium, Cucumber / BDD, JIRA, ISTQB
Prompt engineering	Prompting techniques for agents and RAG

AI / AGENTIC SYSTEMS

Echo-word — agent orchestrator + specialized subagents

Stack: Kotlin · LLM (Claude / GPT / local model) · REST API · web scraping · RAG

- Reusable orchestrator (multi-agent system) that composes specialized subagents for a given goal: some crawl web pages, others collect data via APIs, others learn from the collected data (RAG / embeddings) — shared base, different roles
- Working end-to-end flows built on the same pattern across domains: agentic job-offer discovery and selection, and car-listing search
- Focused on a complete, repeatable agentic workflow and LLM orchestration — task decomposition, routing, result aggregation (ReAct-like pattern) — not on the number of agents

AI Agent — Jira → automated PR

Stack: Kotlin · Spring WebFlux · Llama 3 (Ollama) · GitHub API · JIRA API

- Agent pulls tasks from Jira, analyzes repository context and prepares a ready PR using a local LLM (Llama 3 via Ollama)
- Automatically fetches PRs and tasks, builds a prompt and generates a risk report
- Reactive architecture (application / domain / infrastructure)

EXPERIENCE

Specialist – Software Designer / Developer · PZU

09.2023 – present

- Assistance AI — secure web platform integrating external GenAI (LLM) APIs for PZU employees; delivered in 11 days, ~1,000 users; application recognized in the Rzeczpospolita Cyfrowa 2024 awards
- AI Ambassador (since 05.2025) — shaping the AI-SDLC strategy across PZU IT; co-building an MCP (Model Context Protocol) server compliant with security policy; running GitHub Copilot and AI-assistant workshops
- Agent evaluation and verification — skills for automated PR code review and agent response-quality assessment; AI knowledge-transfer program
- Built a Developer Experience platform measuring the effectiveness of developer tooling across the organization
- Technical Debt (Dept) — system analyzing dependencies across all modules with repository integration; risk scoring and DORA metrics per repository
- CRA (Central Application Registry) — built from scratch a central IT system registry (React.js + Kotlin); automated service onboarding and relationships
- Maintained and evolved microservice architecture (Kotlin), CI/CD pipelines and integrations in an enterprise environment

Junior Developer · PZU

03.2021 – 09.2023

- Extended an automated testing framework (Cucumber + Java) used by a dozen+ teams — cut new-scenario setup time by ~60%
- Designed a test-data generation microservice (Kotlin) — reduced environment prep from hours to minutes
- Maintained a React.js web dashboard for monitoring and mutating test data, used daily by QA and developers
- Integrations via REST API, SQL and RabbitMQ; CI/CD processes in Bamboo

Automated Testing Specialist · PZU Życie SA

08.2017 – 02.2021

- E2E tests (Selenium + Java) covering critical business paths of insurance systems
- Onboarded and trained new QA engineers (4 people)
- Managed the defect backlog in JIRA; analyzed test scenarios with business stakeholders

LANGUAGES

Polish — native · English — technical / working

Interests: generative AI, multi-agent systems, board games.

EDUCATION

IT Technician — Zespół Szkół Elektronicznych i Licealnych, Warsaw